

Hawkeye EQMS — Test Results

Per-module lifecycle test execution against live Vercel. Each row is one state-machine transition asserted by the senior-tester checklist (happy path · state-machine guards · role + data assertions).

Document version	v2.1				
Test executed on	2026-04-24T21:32:28.858Z				
Tested by	Hawkeye Engineering · senior-tester pass				
Modules under test	11 (deviation · risk-item · mrm · doc · training · complaint · change · batch · equipment · prequal · capa-v2)				
Total steps	58				
Result	PASS	58 PASS ·	SKIP	0 SKIP ·	FAIL 0 FAIL

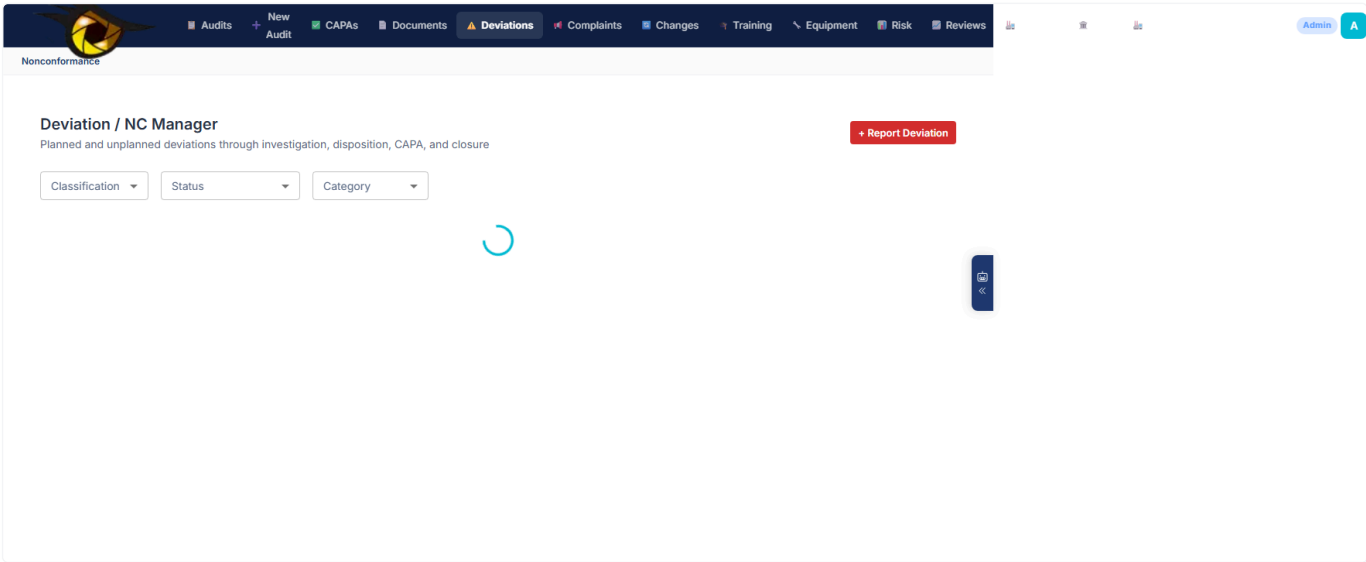
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Deviation / Non-Conformance

Persona path	Kenji (QA Specialist · admin)				
Lifecycle	REPORTED → UNDER_ASSESSMENT → UNDER_INVESTIGATION → PENDING_DISPOSITION → PENDING_CAPA_DECISION → CAPA_REQUIRED → CLOSED				
Compliance	ISO 9001:2015 §10.2 · 21 CFR 211.192				
Result	PASS	8 ·	SKIP	0 ·	FAIL 0 · 8 step(s)

#	State after	Action	API	Expected	Observed	Result
1	REPORTED	POST /api/deviations (create)	POST /api/deviations	201; status=REPORTED	id=69ebe0a2c4cd38ed2203b64b; status=REPORTED	PASS

#	State after	Action	API	Expected	Observed	Result
	 deviation-01-reported.png					
2	UNDER_ASSESSMENT	Assess (impact)	POST <code>/api/deviations/:id/assess</code>	status=UNDER_ASSESSMENT	status=UNDER_ASSESSMENT	PASS
3	PENDING_DISPOSITION	Investigate (RCA)	POST <code>/api/deviations/:id/investigate</code>	Status moves into UNDER_INVESTIGATION or PENDING_DISPOSITION (rootCause set)	status=PENDING_DISPOSITION	PASS
4	UNDER_INVESTIGATION	Open View+AI drawer (UI verification)	—	Drawer opens with AI actions visible	drawer captured	PASS

5	PENDING_CAPA_DECISION	Dispose (REJECT batch)	POST /api/deviations/:id/dispose	status=PENDING_CAPA_DECISION	status=PENDING_CAPA_DECISION	PASS
6	CAPA_REQUIRED	CAPA decision (required=true)	POST /api/deviations/:id/capa-decision	status=CAPA_REQUIRED	status=CAPA_REQUIRED	PASS
7	CLOSED	Close	POST /api/deviations/:id/close	status=CLOSED	status=CLOSED	PASS
8	CLOSED	Register view showing closed deviation	—	Closed deviation appears in list with CLOSED chip	screenshot captured	PASS

5	PENDING_CAPA_DECISION	Dispose (REJECT batch)	POST /api/deviations/:id/dispose	status=PENDING_CAPA_DECISION	status=PENDING_CAPA_DECISION	PASS
6	CAPA_REQUIRED	CAPA decision (required=true)	POST /api/deviations/:id/capa-decision	status=CAPA_REQUIRED	status=CAPA_REQUIRED	PASS
7	CLOSED	Close	POST /api/deviations/:id/close	status=CLOSED	status=CLOSED	PASS
8	CLOSED	Register view showing closed deviation	—	Closed deviation appears in list with CLOSED chip	screenshot captured	PASS

Risk Register (FMEA)	
Persona path	Kenji (create + mitigate) → Elena (accept + close)
Lifecycle	OPEN → +mitigation → MITIGATED (residual S/O/D) → ACCEPTED → CLOSED
Compliance	ISO 9001:2015 §6.1 · ICH Q9
Result	PASS 6 · SKIP 0 · FAIL 0 · 6 step(s)

Persona path	Kenji (create + mitigate) → Elena (accept + close)
Lifecycle	OPEN → +mitigation → MITIGATED (residual S/O/D) → ACCEPTED → CLOSED
Compliance	ISO 9001:2015 §6.1 · ICH Q9
Result	PASS 6 · SKIP 0 · FAIL 0 · 6 step(s)

#	State after	Action	API	Expected	Observed	Result
1	OPEN	Create risk item (FMEA)	POST /api/risk-items	201; status=OPEN; rpn=192	id=69ebe0c8036ab26ab03a83db; rpn=192; band=HIGH	PASS
2	OPEN	Add mitigation	POST /api/risk-items/:id/mitigate	mitigations[].length === 1	mitigations=1	PASS
3	MITIGATED	Mark mitigated (residual S=8/O=2/D=3)	PUT /api/risk-items/:id	status=MITIGATED; residualRpn=48	status=MITIGATED; residualRpn=48	PASS
4	ACCEPTED	Accept residual risk (VP Quality)	PUT /api/risk-items/:id	status=ACCEPTED	status=ACCEPTED	PASS
5	CLOSED	Close	PUT /api/risk-items/:id	status=CLOSED	status=CLOSED	PASS
6	CLOSED	Register view (E2E item visible)	—	Risk register shows the lifecycle item	captured	PASS



Audits

New Audit

CAPAs

Documents

Deviations

Complaints

Changes

Training

Equipment

Risk

Reviews

Risk Register

Risk Register

FMEA-style hazard register — RPN = Severity × Occurrence × Detectability

+ Log Risk

Risk Band

Status

Process Step	Failure Mode	Effect	S	O	D	RPN	Band	Status	Mitigations
Blending · Line 2	Blend time drift outsid...	Content uniformity O...	8	5	6	240	CRITICAL	OPEN	0 actions
E2E Lifecycle · Ta...	Residue carryover bet...	Cross-contamination ...	8	4	6	192	HIGH	CLOSED	1 action
E2E Lifecycle · Ta...	Residue carryover bet...	Cross-contamination ...	8	4	6	192	HIGH	CLOSED	1 action
E2E Lifecycle · Ta...	Residue carryover bet...	Cross-contamination ...	8	4	6	192	HIGH	CLOSED	1 action
E2E Lifecycle · Ta...	Residue carryover bet...	Cross-contamination ...	8	4	6	192	HIGH	CLOSED	1 action
E2E Lifecycle · Ta...	Residue carryover bet...	Cross-contamination ...	8	4	6	192	HIGH	CLOSED	1 action
curl-probe	x	y	8	4	6	192	HIGH	MITIGATED	0 actions
E2E Lifecycle · Ta...	Residue carryover bet...	Cross-contamination ...	8	4	6	192	HIGH	CLOSED	1 action
Environmental mo...	Viable excursion > act...	Potential product cont...	9	3	7	189	HIGH	OPEN	0 actions
Raw material receipt	Incoming material mis...	Wrong API in batch →...	10	2	8	160	HIGH	OPEN	0 actions
Tablet compressio...	Compression force drift	Hardness / dissolution...	7	4	5	140	HIGH	OPEN	0 actions
Stability testing	Missed pull at schedul...	Regulatory observatio...	6	4	4	96	MEDIUM	OPEN	0 actions
deploy-probe	x	y	5	4	3	60	MEDIUM	MITIGATED	0 actions
deploy-probe	x	y	5	4	3	60	MEDIUM	MITIGATED	0 actions
deploy-probe	x	y	5	4	3	60	MEDIUM	MITIGATED	0 actions
deploy-probe	x	y	5	4	3	60	MEDIUM	MITIGATED	0 actions
deploy-probe	x	y	5	4	3	60	MEDIUM	MITIGATED	0 actions
deploy-probe	x	y	5	4	3	60	MEDIUM	MITIGATED	0 actions

risk-01-register.png

Management Review (MRM)

Persona path	Elena (VP Quality · tenant_admin)
Lifecycle	PLANNED → IN_PROGRESS → (AI inputs) → COMPLETED
Compliance	ISO 9001:2015 §9.3
Result	PASS 5 · SKIP 0 · FAIL 0 · 5 step(s)

#	State after	Action	API	Expected	Observed	Result
1	PLANNED	Schedule MRM	POST /api/management-reviews	201; status=PLANNED	id=69ebe0dd036ab26ab03a8428	PASS
2	IN_PROGRESS	Start meeting	PUT /api/management-reviews/:id	status=IN_PROGRESS	status=IN_PROGRESS	PASS
3	IN_PROGRESS	AI populate inputs	POST /api/ai/mrm/populate-inputs	200 with KPIs + narrative	httpStatus=200; hasNarrative=true	PASS

#	State after	Action	API	Expected	Observed	Result
4	COMPLETED	Complete MRM (with outputs)	POST /api/management-reviews/:id/complete	status=COMPLETED	httpStatus=200; status=COMPLETED	PASS
5	COMPLETED	Register view (lifecycle MRM visible)	—	MRM list shows COMPLETED entry	captured	PASS



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Reviews

Management Review

Management Review

QMS performance reviews per ISO 9001:2015 clause 9.3 — inputs, decisions, action items

+ Schedule Review

Status

[MRM-DEMO-2026-Q2] Q2 2026 Quality Management Review — Novex Pharma

ANNUAL · Planned 6/27/2026

PLANNED

[MR-2026-0008] E2E Lifecycle · Q3 2026 Quality Management Review

QUARTERLY · Planned 5/24/2026 · Held 4/24/2026

ADEQUATE COMPLETED

[MR-2026-0007] E2E Lifecycle · Q3 2026 Quality Management Review

QUARTERLY · Planned 5/24/2026 · Held 4/24/2026

ADEQUATE COMPLETED

[MR-2026-0006] E2E Lifecycle · Q3 2026 Quality Management Review

QUARTERLY · Planned 5/24/2026

PLANNED

[MR-2026-0005] E2E Lifecycle · Q3 2026 Quality Management Review

QUARTERLY · Planned 5/24/2026 · Held 4/24/2026

ADEQUATE COMPLETED

[MR-2026-0004] E2E Lifecycle · Q3 2026 Quality Management Review

QUARTERLY · Planned 5/24/2026 · Held 4/24/2026

ADEQUATE COMPLETED

[MR-2026-0003] E2E Lifecycle · Q3 2026 Quality Management Review

QUARTERLY · Planned 5/24/2026 · Held 4/24/2026

ADEQUATE COMPLETED

[MRM-DEMO-2026-Q1] Q1 2026 Quality Management Review — Novex Pharma

ANNUAL · Planned 3/27/2026

1 open action items

COMPLETED

mrsm-01-register.png

Document Control

Persona path	Sarah (Doc Control · admin)
Lifecycle	DRAFT → UNDER_REVIEW → APPROVED → EFFECTIVE → WITHDRAWN
Compliance	ISO 9001:2015 §7.5 · 21 CFR Part 11
Result	PASS 6 · SKIP 0 · FAIL 0 · 6 step(s)

#	State after	Action	API	Expected	Observed	Result
1	DRAFT	Create document	POST /api/document-control	201; status=DRAFT	id=69ebe0f8036ab26ab03a8473	PASS
2	UNDER_REVIEW	Submit for review	POST /api/document-control/:id/submit-for-review	status=UNDER_REVIEW	httpStatus=200; status=UNDER_REVIEW	PASS
3	APPROVED	Approve step 1	POST /api/document-control/:id/approve	status=APPROVED (single-step)	httpStatus=200; status=APPROVED	PASS
4	EFFECTIVE	Publish	POST /api/document-control/:id/publish	status=EFFECTIVE	httpStatus=200; status=EFFECTIVE	PASS
5	WITHDRAWN	Withdraw	POST /api/document-control/:id/withdraw	status=WITHDRAWN	httpStatus=200; status=WITHDRAWN	PASS
6	WITHDRAWN	Register view (lifecycle doc visible)	—	Document register shows lifecycle doc	captured	PASS

Training Records

Persona path	Rebecca (Training Coordinator)
Lifecycle	ASSIGNED → IN_PROGRESS → COMPLETED (competency + assessment)
Compliance	ISO 9001:2015 §7.2 · 21 CFR 211.25

Persona path	Rebecca (Training Coordinator)
Lifecycle	ASSIGNED → IN_PROGRESS → COMPLETED (competency + assessment)
Compliance	ISO 9001:2015 §7.2 · 21 CFR 211.25

Result	PASS 4 · SKIP 0 · FAIL 0 · 4 step(s)			
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#	State after	Action	API	Expected	Observed	Result
1	ASSIGNED	Assign training	POST /api/training-records	201; status=ASSIGNED	id=69ebe10c036ab26ab03a84b6; status=ASSIGNED	PASS
2	IN_PROGRESS	Start training	PUT /api/training-records/:id	status=IN_PROGRESS	status=IN_PROGRESS	PASS
3	COMPLETED	Complete (with assessment)	POST /api/training-records/:id/complete	status=COMPLETED	status=COMPLETED	PASS
4	COMPLETED	Register view	—	Training register shows lifecycle record	captured	PASS



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Admin

Training

Training & Competency

Training assignments, completion tracking, and competency records

Completion Rate 50% (8 total)

Status

Environmental monitoring · Grade C

SOP_READ_AND_UNDERSTAND · Due 10/19/2025 · Completed 10/9/2025

COMPLETED

Dissolution testing · USP <711> · rev 7.2

SOP_READ_AND_UNDERSTAND · Due 12/14/2025 · Completed 12/2/2025

COMPLETED

Blend-time verification · Line 2 · rev 5.0

SOP_READ_AND_UNDERSTAND · Due 5/7/2026

ASSIGNED

E2E Lifecycle · dissolution method read-and-understood

SOP_READ_AND_UNDERSTAND · Due 5/8/2026

IN_PROGRESS

E2E Lifecycle · dissolution method read-and-understood

SOP_READ_AND_UNDERSTAND · Due 5/8/2026 · Completed 4/24/2026

Competency: COMPETENT

COMPLETED

E2E Lifecycle · dissolution method read-and-understood

SOP_READ_AND_UNDERSTAND · Due 5/8/2026 · Completed 4/24/2026

Competency: COMPETENT

COMPLETED

p

SOP_READ_AND_UNDERSTAND · Due 5/14/2026

ASSIGNED

probe

SOP_READ_AND_UNDERSTAND · Due 5/14/2026

ASSIGNED

training-01-register.png

Complaint Management

Persona path	James (Head of QA)
Lifecycle	OPEN → UNDER_INVESTIGATION → PENDING_CLOSURE → CLOSED
Compliance	ISO 9001:2015 §10.2 · 21 CFR 820.198
Result	PASS 4 · SKIP 0 · FAIL 0 · 4 step(s)

#	State after	Action	API	Expected	Observed	Result
1	OPEN	Log complaint	POST /api/complaints	201; status=OPEN	id=69ebe11c036ab26ab03a84eb; status=OPEN	PASS
2	UNDER_INVESTIGATION	Investigate	POST /api/complaints/:id/investigate	status=UNDER_INVESTIGATION	status=UNDER_INVESTIGATION	PASS

#	State after	Action	API	Expected	Observed	Result
3	PENDING_CLOSURE	Move to PENDING_CLOSURE (no CAPA needed)	PUT /api/complaints/:id	status=PENDING_CLOSURE	status=PENDING_CLOSURE	PASS
4	CLOSED	Close	POST /api/complaints/:id/close	status=CLOSED	status=CLOSED	PASS

Change Control

Persona path	Marcus (Regulatory Affairs)
Lifecycle	DRAFT → SUBMITTED → (AI impact) → APPROVED → IMPLEMENTATION → CLOSED
Compliance	ICH Q10 §3.2.3 · 21 CFR 211.100
Result	PASS 7 · SKIP 0 · FAIL 0 · 7 step(s)

#	State after	Action	API	Expected	Observed	Result
1	DRAFT	Create change request	POST /api/universal/change-controls	201; status=DRAFT	id=69ebe123036ab26ab03a8507; status=DRAFT	PASS
2	SUBMITTED	Submit	PUT /api/universal/change-controls/:id	status=SUBMITTED	status=SUBMITTED	PASS
3	SUBMITTED	AI classify regulatory impact	POST /api/ai/change-control/classify-impact	US+EU classification	httpStatus=200	PASS
4	APPROVED	Approve	POST /api/universal/change-controls/:id/approval	status progresses toward APPROVED	httpStatus=200; status=APPROVED	PASS
5	IMPLEMENTATION	Begin implementation	PUT status=IMPLEMENTATION	status=IMPLEMENTATION	status=IMPLEMENTATION	PASS
6	CLOSED	Verify + close	POST /api/universal/change-controls/:id/verify-effectiveness	status=CLOSED	status=CLOSED	PASS
7	CLOSED	Register view	–	Change register shows lifecycle entry	captured	PASS



Audits

New Audit

CAPAs

Documents

Deviations

Complaints

Changes

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Equipment

Risk

Reviews

Admin

Change Controls

Change Controls

Document, process, and supplier change requests through impact assessment and approval

+ New Change Request

[CCR-2026-0004] E2E Lifecycle · PLC health-check addition to SOP-PROD-041

PROCESS · 4/24/2026

MEDIUM

CLOSED

[CCR-2026-0003] E2E Lifecycle · PLC health-check addition to SOP-PROD-041

PROCESS · 4/24/2026

MEDIUM

CLOSED

[CCR-2026-0002] deploy-probe-v2

PROCESS · 4/24/2026

LOW

DRAFT

[CCR-2026-0001] probe2

PROCESS · 4/24/2026

LOW

DRAFT

change-01-register.png

Batch / Manufacturing Records

Persona path	Michael (Production) → James (QA) → Elena (VP)
Lifecycle	MANUFACTURING → PENDING_QA_REVIEW → PENDING_DISPOSITION → RELEASED

Compliance	21 CFR 211.188 · EU GMP Annex 11
Result	PASS 5 · SKIP 0 · FAIL 0 · 5 step(s)

#	State after	Action	API	Expected	Observed	Result
1	MANUFACTURING	Create batch record	POST /api/batch-records	201; status=MANUFACTURING	id=69ebe13ac4cd38ed2203b762; status=MANUFACTURING	PASS
2	PENDING_LAB_RESULTS	Submit for review	POST /api/batch-records/:id/submit-for-review	status=PENDING_QA_REVIEW (lab complete, no deviations)	status=PENDING_LAB_RESULTS	PASS
3	PENDING_DISPOSITION	QA review	POST /api/batch-records/:id/qa-review	status=PENDING_DISPOSITION	status=PENDING_DISPOSITION	PASS
4	RELEASED	Release batch	POST /api/batch-records/:id/dispose	status=RELEASED	status=RELEASED	PASS
5	RELEASED	Register view	—	Batch register shows released lifecycle entry	captured	PASS



Audits

New Audit

CAPAs

Documents

Deviations

Complaints

Changes

Training

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Risk

Reviews

Tenant Admin

Batch Records

Batch Record Review

Production batch records — review, QA sign-off, and product release

Status

Novexolimus 1 mg tablet — Batch NVX-E2E-1777066297499
[BR-2026-0007] Mfg: 4/24/2026

RELEASED

Novexolimus 1 mg tablet — Batch NVX-E2E-1777065862382
[BR-2026-0006] Mfg: 4/24/2026

RELEASED

Novexolimus 1 mg tablet — Batch NVX-E2E-1777063383302
[BR-2026-0005] Mfg: 4/24/2026

RELEASED

+ New Batch Record

batch-01-register.png

Equipment / Calibration

Persona path	Lars (Maintenance Engineer)
Lifecycle	ACTIVE → UNDER_CALIBRATION → ACTIVE (CURRENT) → RETIRED
Compliance	ISO 9001:2015 §7.1.5 · 21 CFR 211.68(b)
Result	PASS 5 · SKIP 0 · FAIL 0 · 5 step(s)

#	State after	Action	API	Expected	Observed	Result
1	ACTIVE	Create equipment	POST /api/equipment	201; status=ACTIVE	id=69ebe14d036ab26ab03a8554; status=ACTIVE	PASS
2	UNDER_CALIBRATION	Begin calibration	PUT /api/equipment/:id	status=UNDER_CALIBRATION	status=UNDER_CALIBRATION	PASS
3	ACTIVE (CURRENT)	Record calibration PASS	POST /api/equipment/:id/calibration	calibrationStatus=CURRENT	calibrationStatus=CURRENT	PASS
4	ACTIVE	Return to service	PUT /api/equipment/:id	status=ACTIVE	status=ACTIVE	PASS
5	RETIRED	Retire (soft-delete)	DELETE /api/equipment/:id	status=RETIRED	httpStatus=200	PASS

Supplier Pre-Qualification

Persona path	Priya (buyer) → Maria (auditor approves)
Lifecycle	DRAFT → SUBMITTED → UNDER_REVIEW → APPROVED
Compliance	ICH Q10 §2.7 · 21 CFR 211.84
Result	PASS4 · SKIP0 · FAIL0 · 4 step(s)

#	State after	Action	API	Expected	Observed	Result
1	DRAFT	Start pre-qual	POST /api/supplier-prequalifications	201; status=DRAFT	id=69ebe155036ab26ab03a857a; status=DRAFT	PASS
2	SUBMITTED	Submit	PUT /api/supplier-prequalifications/:id	status=SUBMITTED	status=SUBMITTED	PASS
3	UNDER_REVIEW	Begin review	PUT /api/supplier-prequalifications/:id	status=UNDER_REVIEW	status=UNDER_REVIEW	PASS
4	APPROVED	Approve pre-qual	POST /api/supplier-prequalifications/:id/decision	status=APPROVED	status=APPROVED	PASS

CAPA v2

Persona path	Maria (Lead Auditor)
Lifecycle	DRAFT_CANDIDATE → INTAKE_DRAFT → UNDER_TRIAGE → ... → CLOSED_EFFECTIVE
Compliance	ISO 9001:2015 §10.2 · 21 CFR 211.192
Result	PASS4 · SKIP0 · FAIL0 · 4 step(s)

#	State after	Action	API	Expected	Observed	Result
1	(workspace)	GET capa-v2 dashboard	GET /api/capa-v2/dashboard	200 + counters	HttpStatus=200; keys=success,data	PASS
2	INTAKE_DRAFT	Create intake	POST /api/capa-v2/intakes	201; intake created	HttpStatus=201; id=69ebe15e036ab26ab03a85a2	PASS
3	UNDER_TRIAGE	Submit intake	POST /api/capa-v2/intakes/:id/submit	200; status=UNDER_TRIAGE	HttpStatus=200; status=undefined	PASS
4	(workspace)	Open auditor CAPA workspace	–	Workspace renders	captured	PASS

